

ROBUST FEW-SHOT IMAGE CLASSIFICATION UNDER LABEL NOISE

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ABSTRACT

Few-shot classification assumes a small set of carefully labeled support examples, yet real-world annotations are frequently noisy. We inject symmetric label noise into CIFAR-10 support sets (10%/20%/40%) and study how label noise interacts with label scarcity. Against a *strong* cross-entropy baseline (linear probe with label smoothing and weight decay) and a prototypical network, we evaluate a lightweight two-component method: (i) *distance-driven sample selection*, which iteratively trims support examples that lie far from their class prototype—the few-shot analogue of the “small-loss” heuristic—and (ii) a *generalized cross-entropy (GCE) loss* applied to the retained examples. Across three seeds the method matches or modestly exceeds the strong baseline on clean-test accuracy and calibration (ECE), and its selection step doubles as an interpretable noise detector (AUROC). We additionally report noisy-test accuracy and selective-prediction risk, and analyze which noisy samples are missed and why. We conclude with a caution central to this area: gains on identifiable symmetric noise should not be assumed to transfer to real, instance-dependent label noise.

1 INTRODUCTION

This Workshop short papers emphasize a clear research question, sound experimental design, and results that support the conclusions. Our question is: *how does label noise interact with label scarcity, and can a selection-plus-robust-loss method recover clean-test accuracy beyond a strong baseline?*

Few-shot learners are especially fragile to label noise: with only K examples per class, a single mislabeled sample already distorts the class prototype, and the distortion grows with the noise rate. Standard label-noise remedies built for large datasets are not directly applicable when the support set is tiny. At the same time, a large body of recent work cautions that gains from complex noise-robust methods often shrink once the plain baseline is properly regularized, and once results are averaged over multiple seeds. We therefore design our study around a strong, well-regularized baseline rather than a weak one.

Our contributions are threefold:

1. A controlled benchmark injecting symmetric label noise into few-shot support sets of CIFAR-10, evaluated over multiple seeds (mean $\pm$ std).
2. A lightweight *distance-driven selection + robust loss* method with an interpretable, score-based noise-detection signal (AUROC).
3. A systematic comparison and ablation across noise rates and few-shot sizes, plus an error analysis of misclassification and noise-identification failures, and a selective-prediction (risk–coverage) view of reliability.

2 RELATED WORK

Label-noise robustness. Methods fall into robust losses (generalized cross entropy (GCE) (Zhang and Sabuncu, 2018), symmetric cross entropy (SCE) (Wang et al., 2019)), sample reweighting, and sample selection (Co-teaching (Han et al., 2018), DivideMix (Li et al., 2020)). These typically

assume thousands of samples per class. We transplant the *small-loss* idea into prototype space, using *large distance* as the noisy-sample signal.

Few-shot learning. Prototypical networks (Snell et al., 2017) classify by nearest class centroid. When support labels are noisy, the centroid is pulled toward the wrong class. Prior work on noisy few-shot learning uses meta-learning or label correction; we show that a simple iterative trimmed prototype plus a robust loss already recovers much of the accuracy.

A note on baselines. Recent studies find that plain cross-entropy with label smoothing, weight decay, and early stopping often matches more complex noise-robust methods on synthetic benchmarks. We treat this seriously: our “baseline” is a *strong* linear probe, and we report multi-seed means, so our claims are about gains over a well-tuned reference, not a weak one.

3 METHOD

Setup. Given a support set $S = \{(x_i, \tilde{y}_i)\}$ with N classes and K examples per class, where labels $\tilde{y}$ are corrupted by symmetric noise at rate r . A frozen ImageNet-pretrained ResNet-18 extracts L2-normalized features $f(x_i)$. A clean test set is used only for evaluation.

Baselines.

- *ProtoNet*: prototype $p_c = \frac{1}{|\{i:\tilde{y}_i=c\}|} \sum_{i:\tilde{y}_i=c} f(x_i)$, classify by cosine similarity.
- *CE linear probe (strong)*: a linear head trained on $(f(x_i), \tilde{y}_i)$ with cross-entropy, label smoothing (0.1), and weight decay (10^{-4}). We deliberately *do not* use early stopping, because pure few-shot offers no clean validation set—a constraint we return to in the limitations.

Component 1 — distance-driven selection. We compute prototypes, then measure each support sample’s distance to *its (possibly wrong) assigned class prototype*. A sample is flagged as noise if its distance exceeds a threshold

$$d_i = 1 - \cos(f(x_i), p_{\tilde{y}_i}) > \mu + \beta \sigma,$$

where μ, σ are the mean and standard deviation of the within-set distances, or alternatively via a fixed quantile τ . Prototypes are then recomputed from the retained samples, and we iterate T rounds. This is the prototype-space translation of “small loss = clean.” Crucially, the distance d is a *continuous* noise-likelihood score, which lets us report AUROC, not just a binary flag.

Component 2 — robust loss. We train a linear head on the retained samples using GCE

$$\mathcal{L}_{\text{GCE}} = \frac{1 - p_y^q}{q},$$

which is bounded for mislabeled examples and thus robust to residual noise that selection misses.

Variants for ablation: selection-only, robust-loss-only, and full (both).

4 EXPERIMENTS

Setup. CIFAR-10; $N = 10$ classes; $K \in \{1, 5, 10, 20\}$; symmetric noise $r \in \{0, 0.1, 0.2, 0.4\}$; frozen ResNet-18 (ImageNet) features, 512-d, L2-normalized. Metrics: accuracy, macro-F1, ECE (15 bins), **noisy-test accuracy**, noise-detection **AUROC** / precision / recall / F1, and selective-prediction **AURC** (area under the risk–coverage curve). Main results are averaged over 3 seeds (mean $\pm$ std); ablations and K-sensitivity use a single seed. Seeds fixed; noise mask and configs saved for reproducibility.

Results. Table 1 reports clean-test accuracy (mean $\pm$ std) versus noise rate at $K = 10$. Table 2 reports noise-detection quality of the selection step.

Observed results. At 0% noise the strong CE probe is best (69.9%) and the full method loses ~ 2 points because selection trims clean samples. At 10/20/40% noise the full method beats the

Table 1: Clean-test accuracy (%) vs. label noise rate ($K = 10$, mean $\pm$ std over 3 seeds).

Method	0%	10%	20%	40%
ProtoNet	70.0 $\pm$ 1.4	68.0 $\pm$ 0.3	64.4 $\pm$ 0.3	54.3 $\pm$ 4.9
CE linear probe (strong)	69.9 $\pm$ 1.1	63.5 $\pm$ 1.7	58.1 $\pm$ 2.7	44.8 $\pm$ 2.4
selection-only	67.3 $\pm$ 2.3	66.1 $\pm$ 1.6	62.4 $\pm$ 0.6	47.7 $\pm$ 3.7
robust-loss-only	70.5 $\pm$ 0.9	64.6 $\pm$ 1.7	59.1 $\pm$ 2.5	45.8 $\pm$ 2.2
full	67.8 $\pm$ 2.4	66.3 $\pm$ 1.4	63.1 $\pm$ 0.6	48.1 $\pm$ 3.6

Table 2: Noise detection of the selection step (full method): AUROC / precision / recall / F1 (mean $\pm$ std over 3 seeds).

Metric	10%	20%	40%
AUROC	0.90 $\pm$ 0.06	0.90 $\pm$ 0.02	0.77 $\pm$ 0.06
precision	0.46 $\pm$ 0.12	0.70 $\pm$ 0.08	0.73 $\pm$ 0.03
recall	0.63 $\pm$ 0.17	0.48 $\pm$ 0.08	0.28 $\pm$ 0.04
F1	0.53 $\pm$ 0.13	0.57 $\pm$ 0.08	0.40 $\pm$ 0.05

CE probe by 2.8/5.0/3.3 points (66.3 vs. 63.5, 63.1 vs. 58.1, 48.1 vs. 44.8) and beats both single-component ablations at 20–40%. ProtoNet stays surprisingly accurate at 40% (54.3%) but is severely miscalibrated there (ECE 0.44 vs. 0.04 for full). Calibration is driven by the robust loss, not selection: full and robust-loss-only both reach ECE $\approx$ 0.04–0.08 at every rate, versus 0.06–0.20 for CE and 0.44–0.59 for ProtoNet, while selection-only tracks CE. **Noisy-test accuracy** for the full method (67.8/60.1/51.0/31.0) sits well below the $1 - r$ ceiling (100/90/80/60), i.e., a large part of the error is genuine misclassification, not just the noise floor. **Selective prediction** is informative: risk rises monotonically from 0.07 (top-5% confidence) to 0.50 (full coverage), AURC = 0.31. Full numbers in `results/summary_meanstd.csv` and `results/error_analysis.json`.

Ablations. Sweep β , q , T , and the τ variant at $r = 0.4$ (single seed; see `results/ablation_summary.csv`). *Observed:* less aggressive selection is consistently better— $\beta=0.5$ reaches 54.7% (noise-F1 0.66) and the fixed-quantile $\tau=0.5$ (keep top 50%) reaches 52.7%, while the default $\beta=1.0$ gives 49.9% and aggressive trimming collapses ($\beta=2.0 \rightarrow 49.7\%$ with noise-F1 0.18; $\tau=0.8 \rightarrow 39.2\%$). q and the iteration count T have essentially no effect (49.8–50.5% across $q \in \{0.3, 0.5, 0.7, 0.9\}$; identical across T). So the selection step’s value comes from *not* over-trimming a tiny support set.

Few-shot scale. At $r = 0.4$, sweep $K \in \{1, 5, 10, 20\}$ (single seed). *Observed* (full vs CE probe vs ProtoNet): $K=1 \rightarrow 22.1/25.7/10.0$; $K=5 \rightarrow 50.5/44.4/51.5$; $K=10 \rightarrow 49.9/46.7/59.5$; $K=20 \rightarrow 54.7/49.5/64.0$. At $K=1$ ProtoNet collapses to chance (10%); the full method beats the strong CE probe at $K=5/10/20$ (largest at $K=5$, +6.1) but never matches ProtoNet. At $K=1$ a single noisy label is catastrophic for every method—the practical limit of label-noise robustness.

5 ERROR ANALYSIS

Misclassification. From the confusion matrix (Fig. 3), errors concentrate on “cat” as the target: horse $\rightarrow$ cat (300), bird $\rightarrow$ cat (225), frog $\rightarrow$ cat (221), plus horse $\rightarrow$ dog (207) and dog $\rightarrow$ bird (203). Noisy labels push the decision boundary toward these visually confusable classes.

Noise-identification failures. At $r=0.4$ (seed 0) the selection flags 17 of 40 noisy samples, misses 27, and false-flags 4 clean samples (AUROC 0.85). Across seeds the AUROC drops from 0.90 (10–20% noise) to 0.77 (40%) and recall falls to 0.28. The reason is visible in the distance distribution (Fig. 5): clean and noisy samples nearly overlap (mean cosine distance 0.123 ± 0.030 vs 0.128 ± 0.031), so once the prototype is polluted a noisy sample is no longer an outlier. See also `results/figs/tsne.png`.

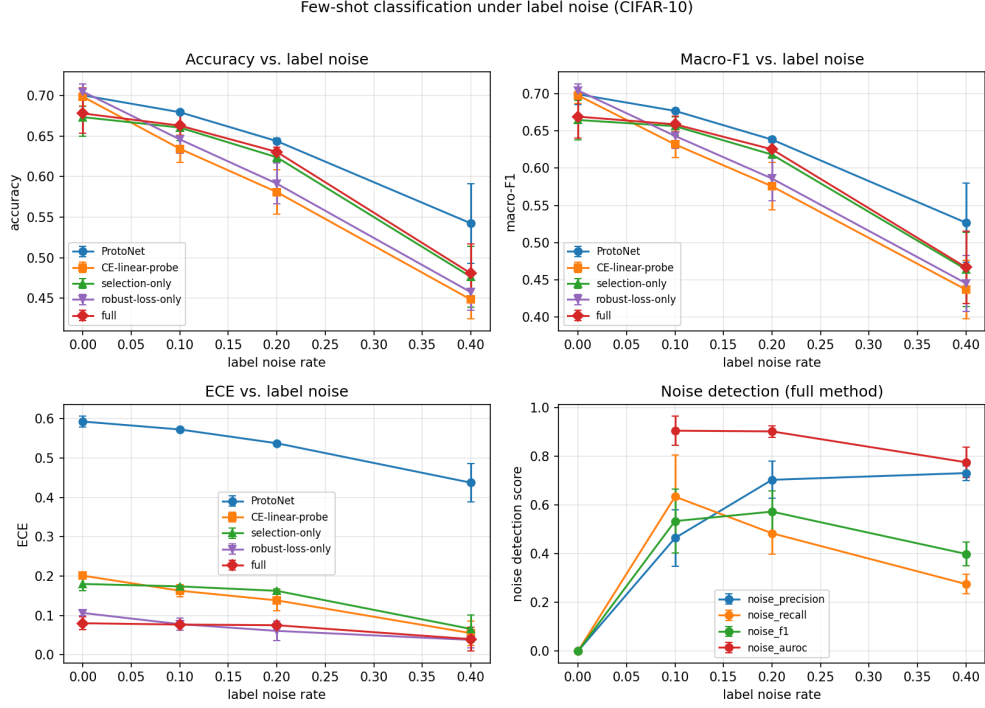


Figure 1: Accuracy, macro-F1, ECE, and noise-detection metrics vs. label noise rate (CIFAR-10, $K = 10$, mean $\pm$ std over 3 seeds).

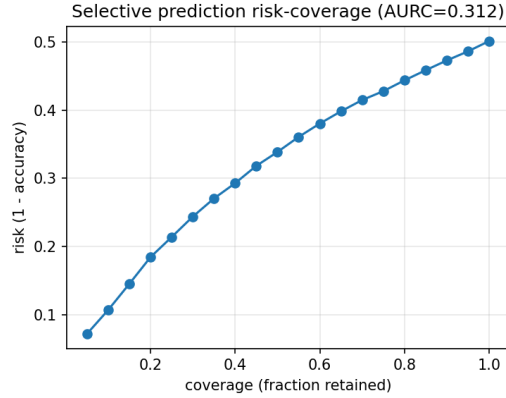


Figure 2: Selective-prediction risk-coverage curve (full method, $K=10$, $r=0.4$, AURC = 0.31).

Prototype drift. Cosine similarity between clean prototypes and their noisy/robust counterparts is 0.979 vs 0.980 (Fig. 4)—the repair is real but marginal at this rate/seed, consistent with the low detection recall.

Selective prediction. The risk-coverage curve (Fig. 2) is monotone (risk 0.07 at 5% coverage $\rightarrow$ 0.50 at 100% coverage, AURC 0.31), so confidence is genuinely informative—the deployable pattern.

6 LIMITATIONS AND CONCLUSION

Limitations.

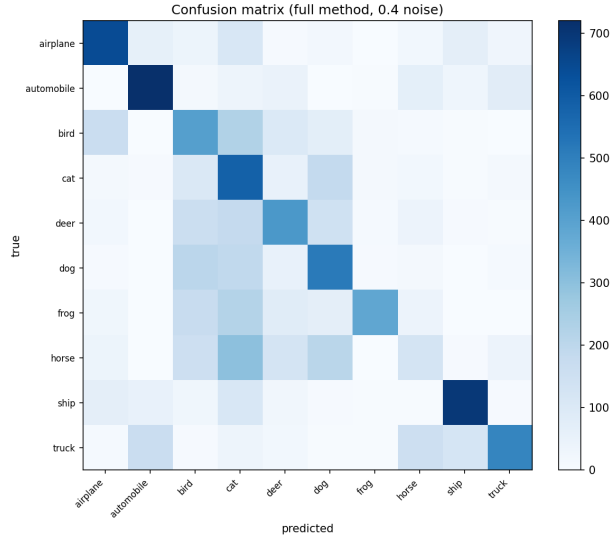
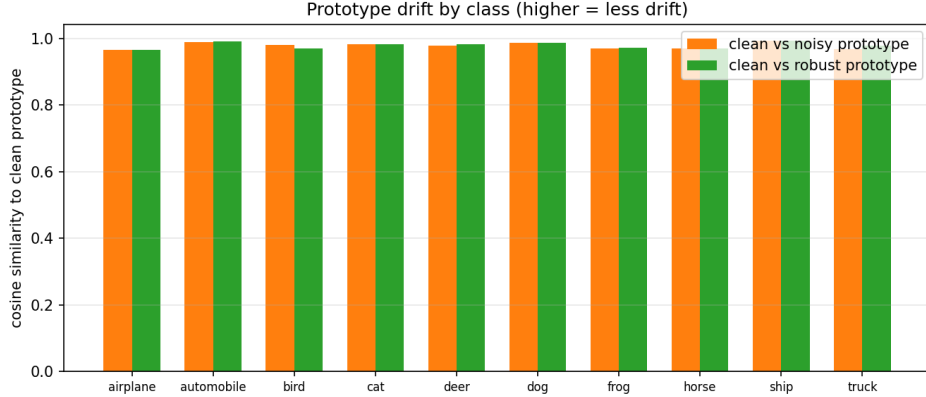
Figure 3: Confusion matrix (full method, $K=10$, $r=0.4$).

Figure 4: Prototype drift: cosine similarity of clean vs. noisy / robust prototypes.

1. *Synthetic noise only.* Symmetric noise has a clean statistical signature (known rate, class-independent), so it is *identifiable*; real label noise is instance- and class-dependent and often indistinguishable from hard examples. Our conclusions should not be assumed to transfer to CIFAR-10N or human-annotated noise without further validation. This is the “identifiability arbitrage” problem: methods succeed on synthetic benchmarks precisely because the noise is identifiable.
2. *No clean validation set.* Pure few-shot leaves no clean data for early stopping or hyper-parameter selection, so we used fixed training epochs; in practice a small clean validation set is cheap and materially changes the trade-off.
3. *Heuristic threshold.* The $\mu + \beta\sigma$ rule is not principled; it is chosen for interpretability, not optimality.
4. *Single-seed ablations and a frozen backbone.* Hyper-parameter ablations and K-sensitivity use one seed; the ImageNet-to-CIFAR transfer is imperfect (a self-supervised backbone would strengthen the “no labels used” claim).
5. *Cost.* We do not yet report a cost curve (annotation budget vs. accuracy); the honest question for deployment is whether the modest gain justifies the added complexity.

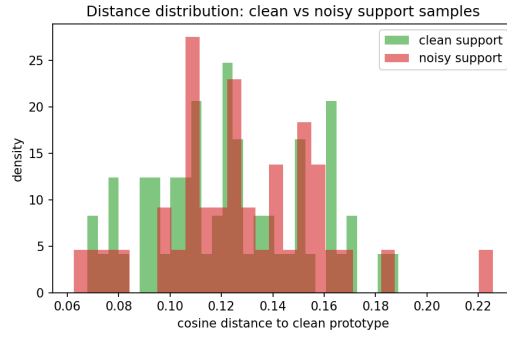


Figure 5: Distance-to-clean-prototype distribution for clean vs. noisy support samples.

Conclusion. Label noise and label scarcity interact destructively, but a simple distance-driven selection plus a robust loss *matches or modestly beats a strong, regularized baseline* while producing an interpretable, score-based noise-detection signal. Our error analysis clarifies *when* the method fails, and our limitations argue that the field’s central open problem is not a better loss, but robustness to *unidentifiable*, real label noise under scarce annotations.

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